

Переосмысление хранения корпоративных данных

с **QSAN**

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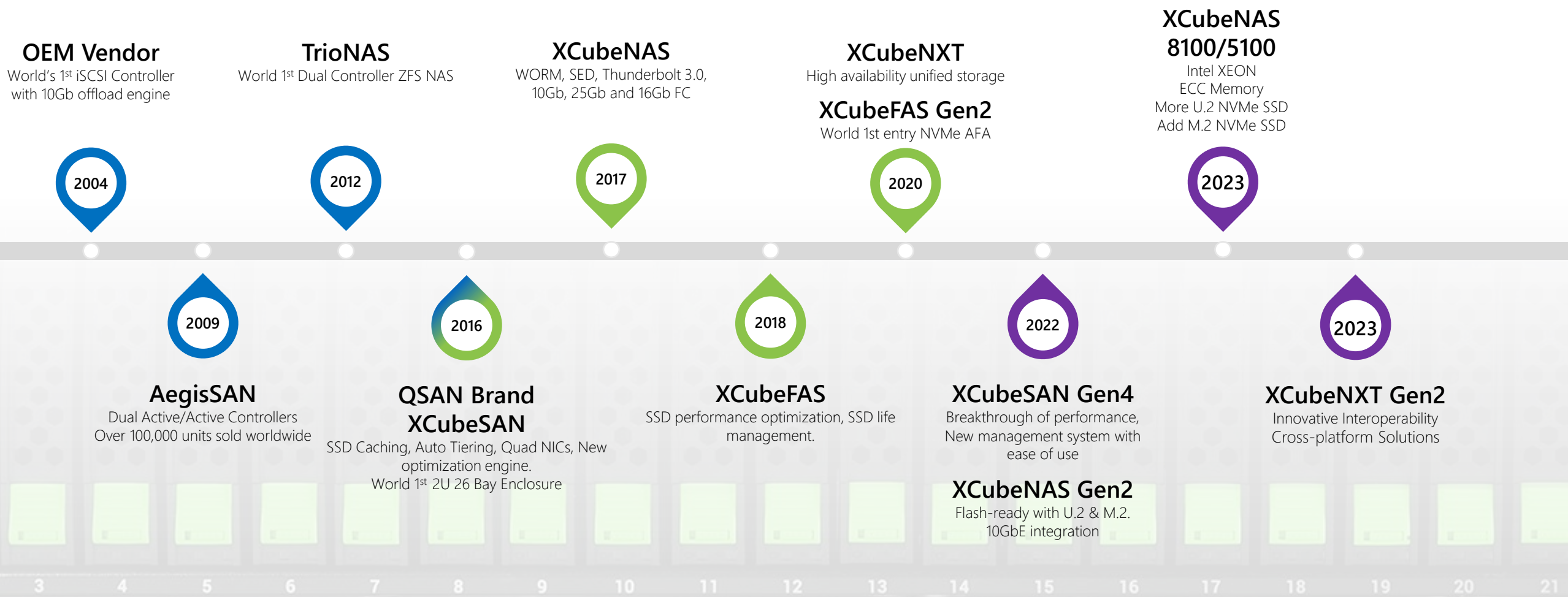
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QSAN

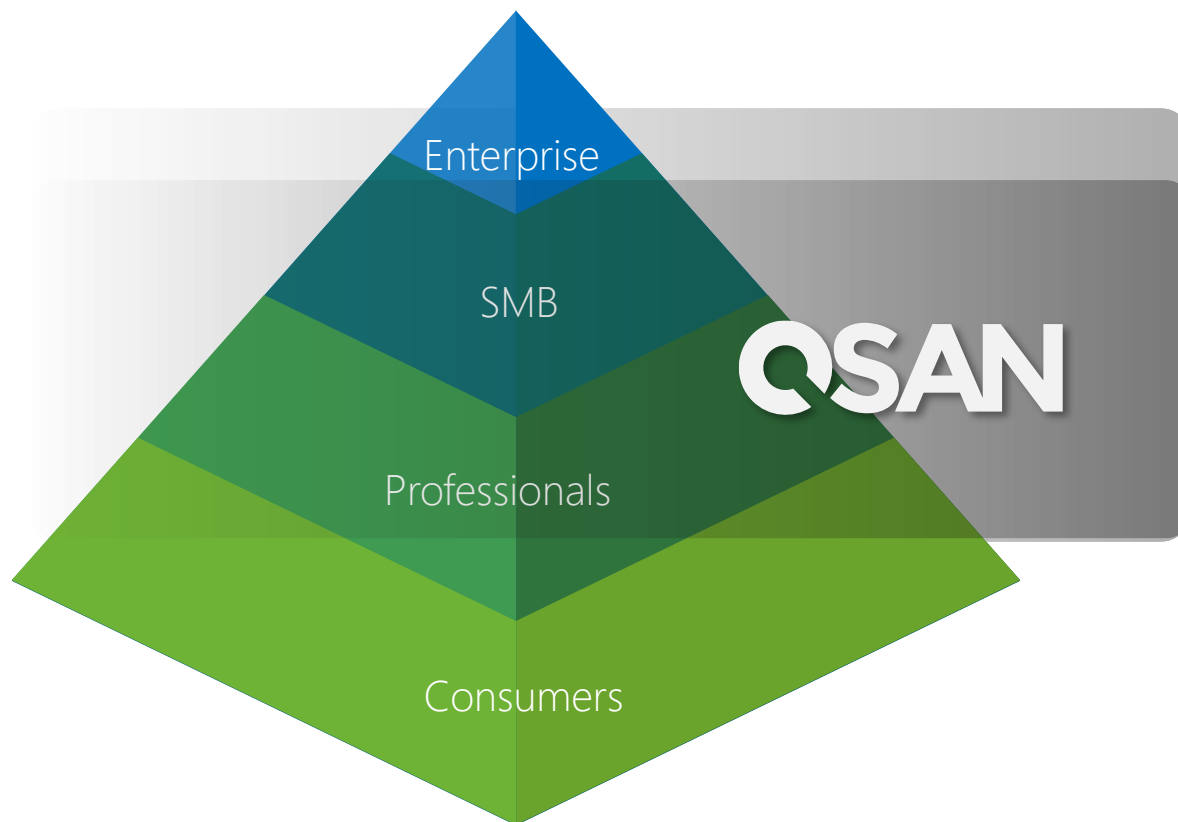
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История QSAN



Позиционирование в маркетинге

СХД корпоративного класса, лидирующая по соотношению
цена/производительность



Bundled with
QSAN
NVMe SSD

XCubeFAS
All-Flash NVMe



XCubeNXT
Unified Storage



XCubeSAN
Hybrid flash



EXPANDING ECOSYSTEM PARTNERS

Connectivity



Drive and Expansion



Virtualization



Backup



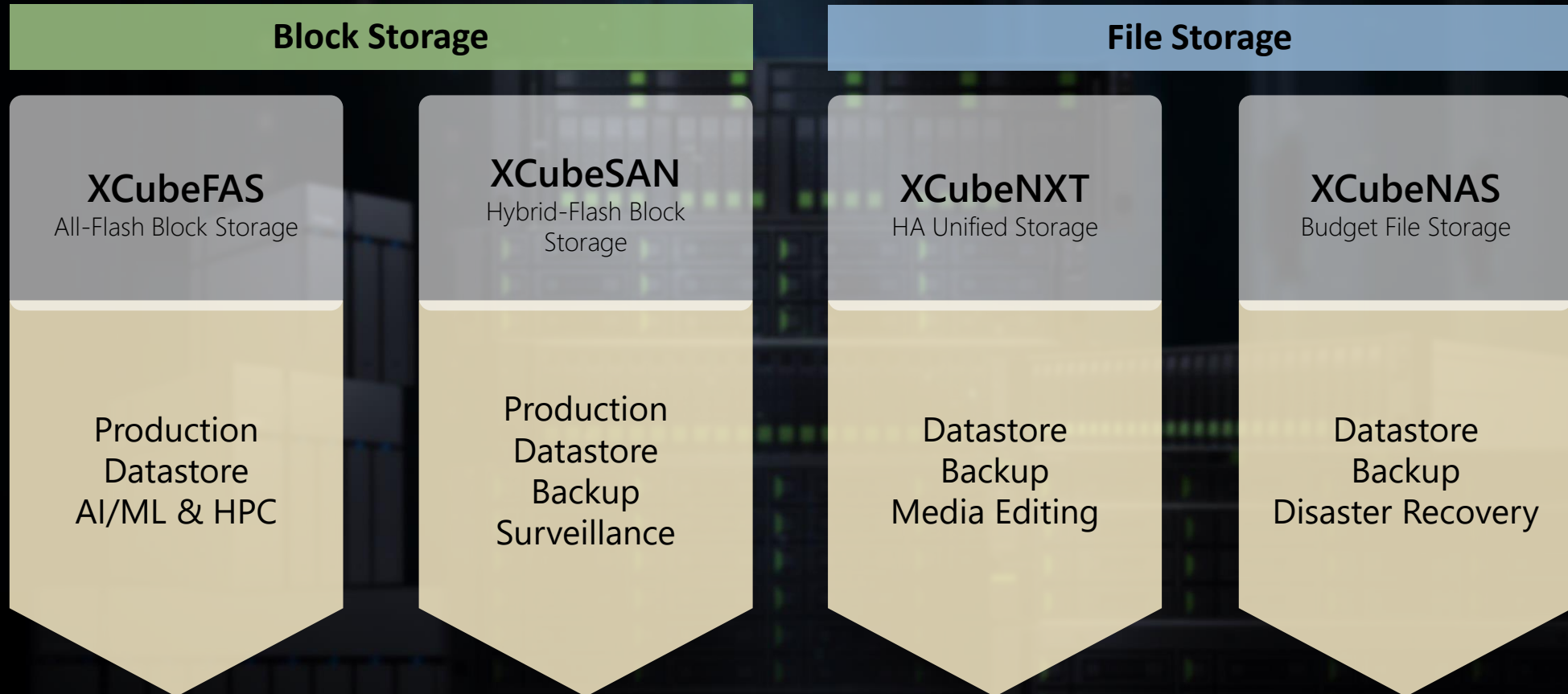
Surveillance



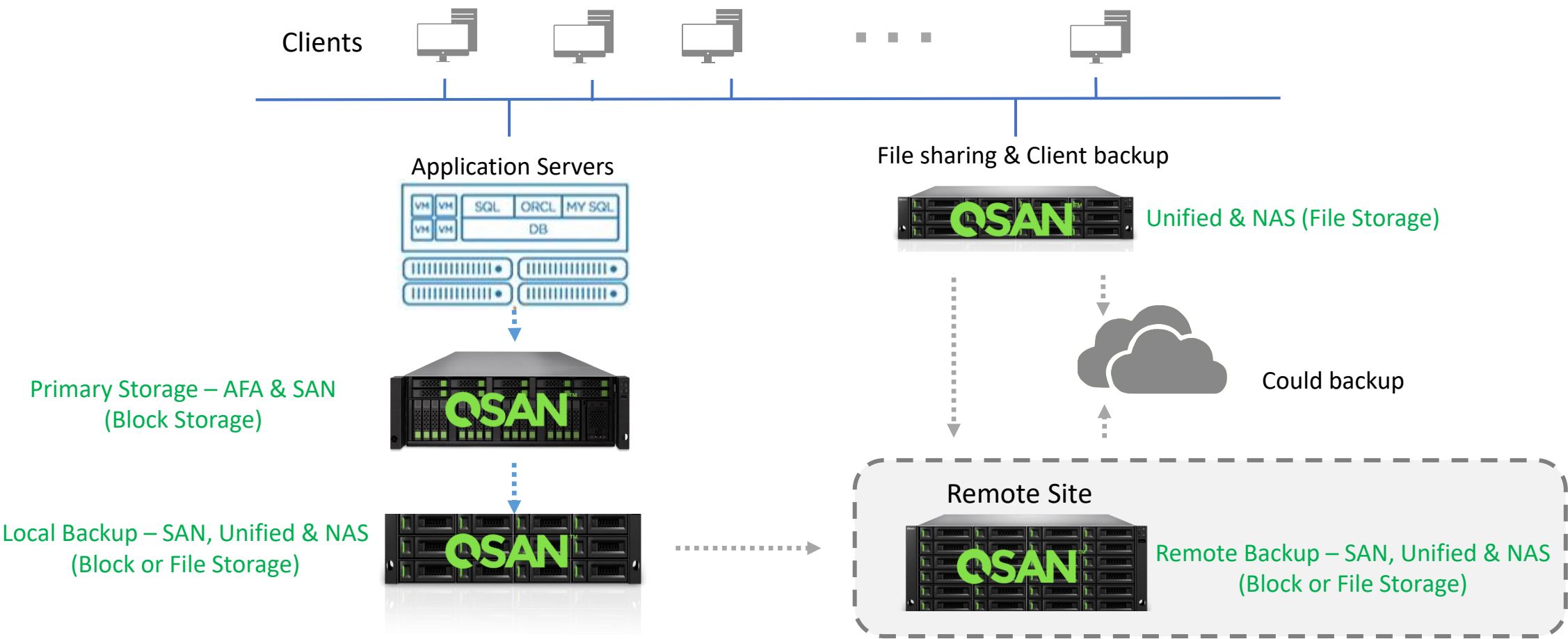
M & E



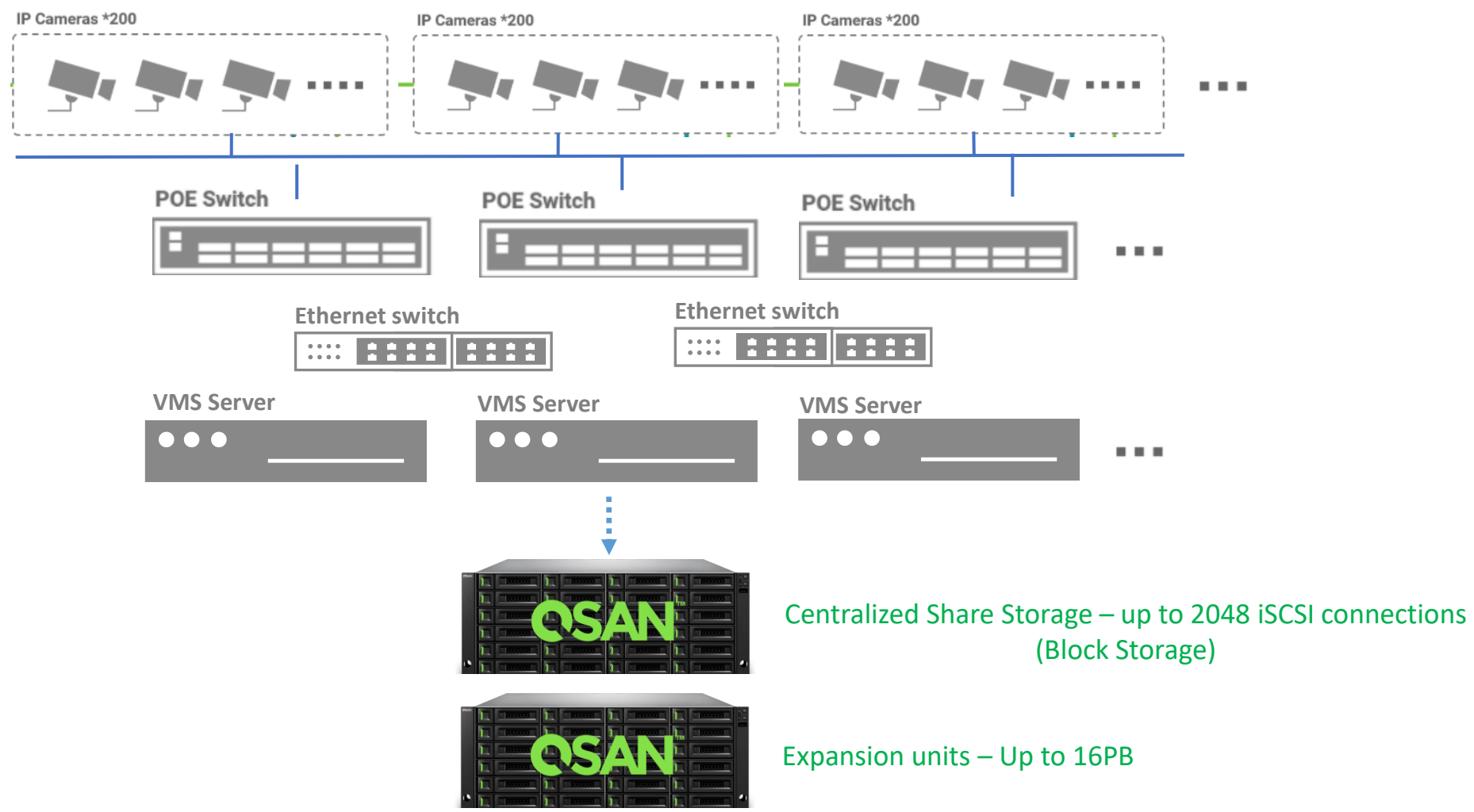
XCube Platform – One Architecture



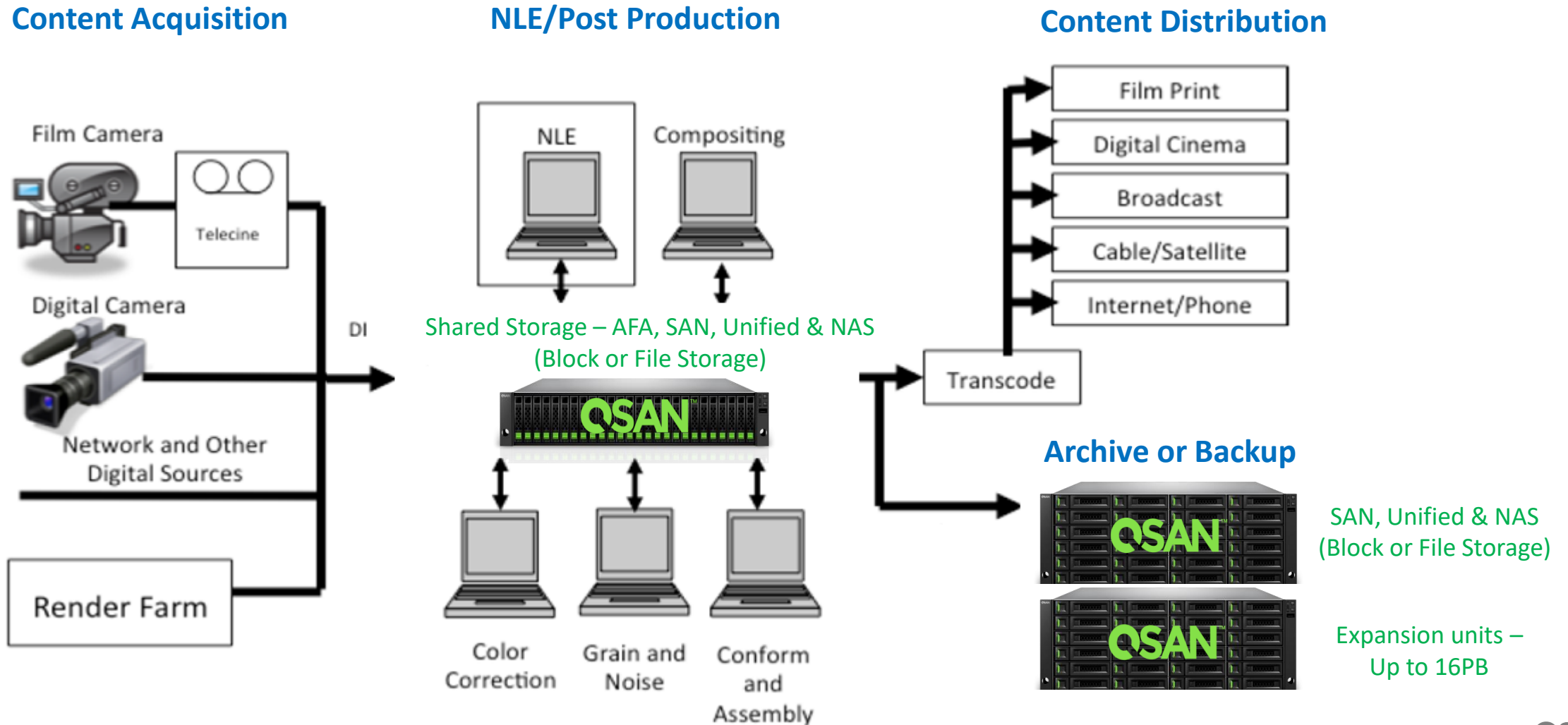
Target Application – Enterprise IT



Target Application – Large Surveillance

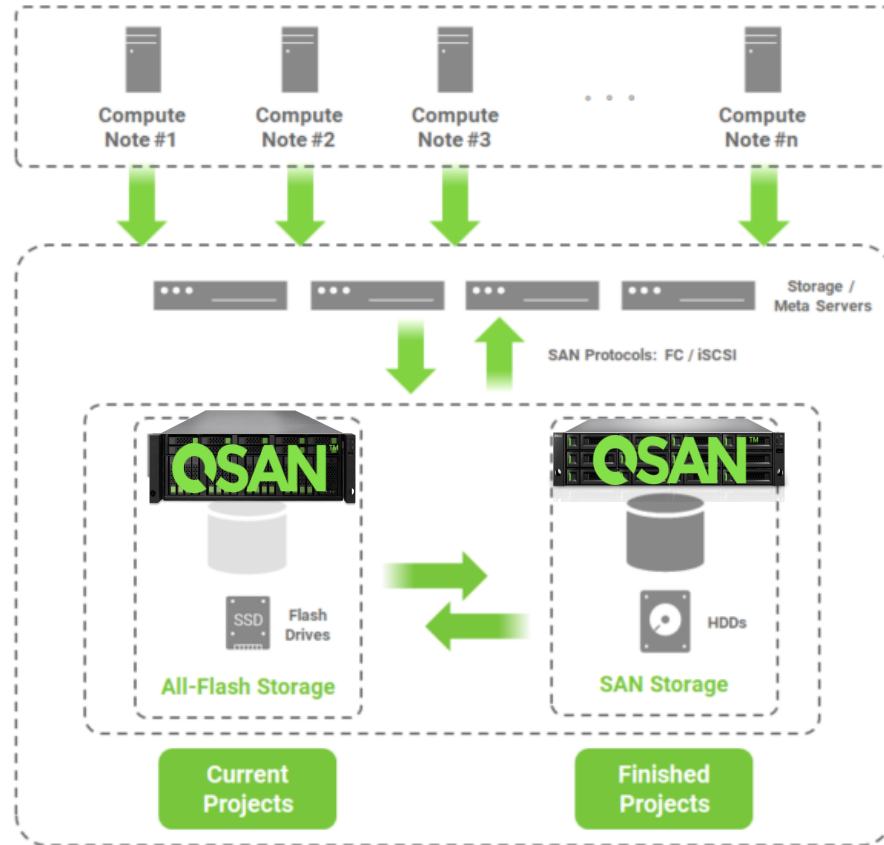


Target Application – Media Editing

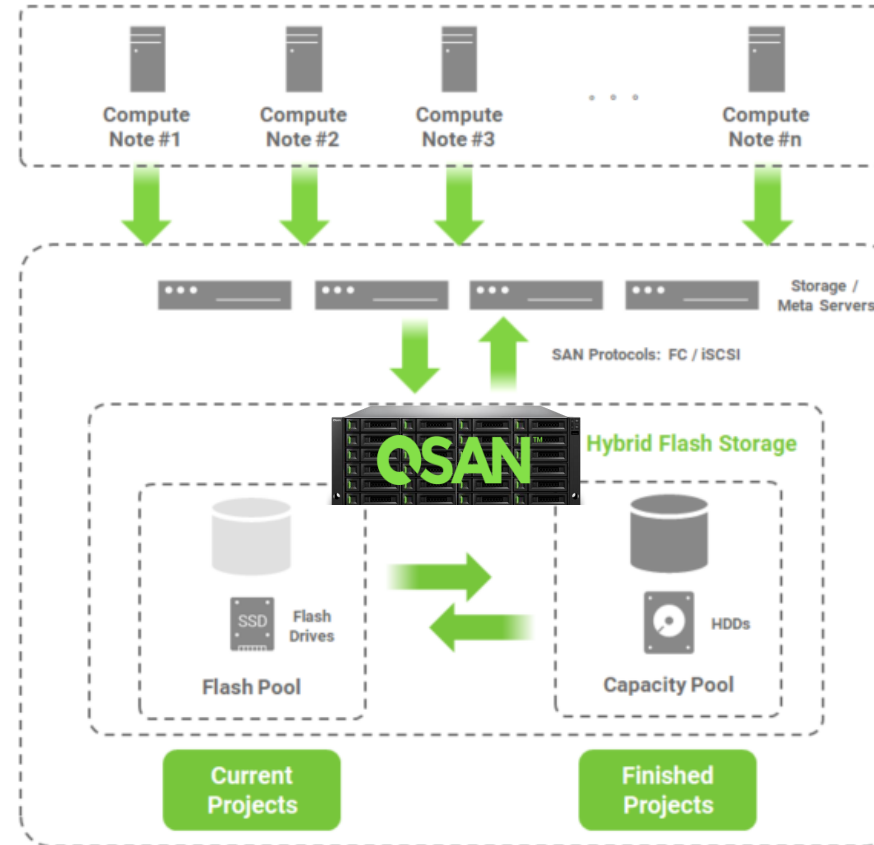


Target Application – HPC & AI

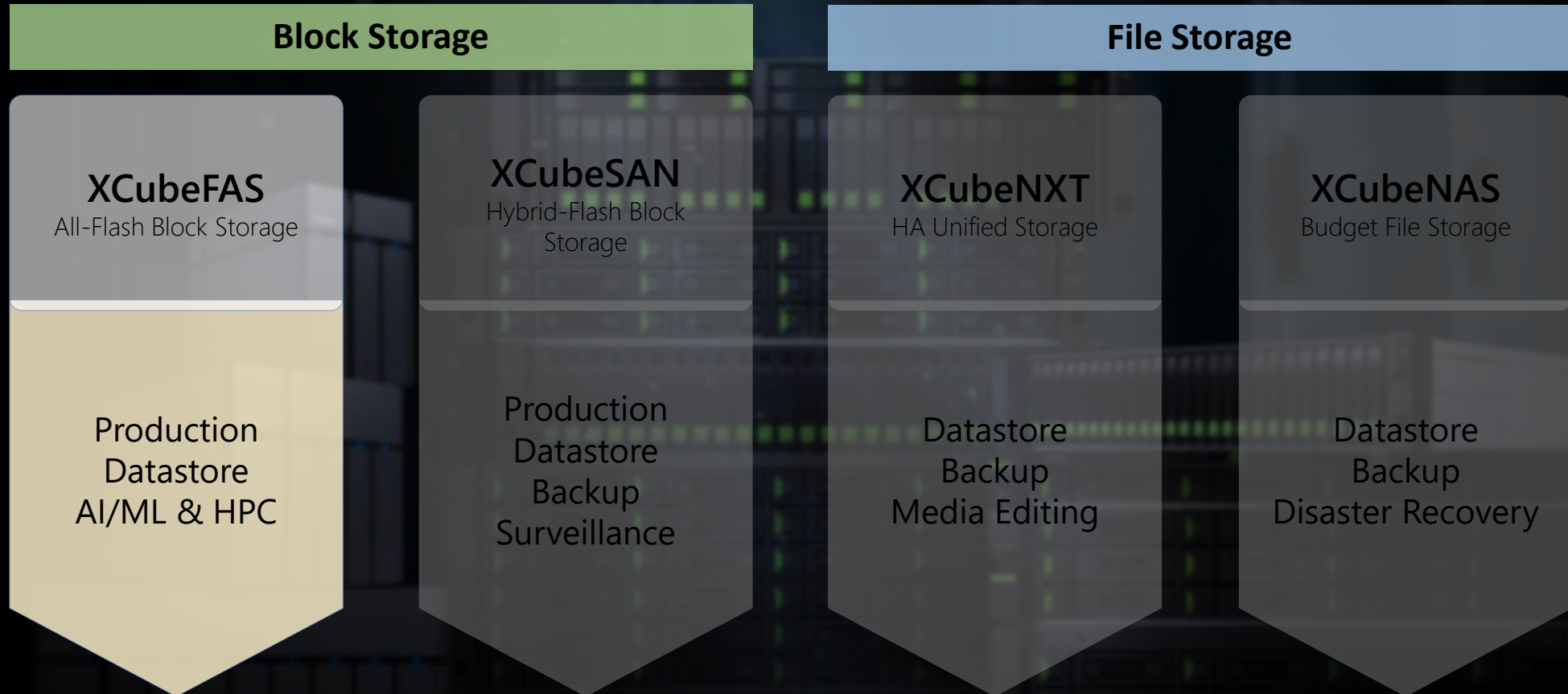
Performance Optimized



Cost Optimized



XCube Platform – One Architecture



QSAN Solution - XCubeFAS XF3126D



XCubeFAS Gen2 3U26

Dual-Port NVMe SSDs

Random Write Performance

110,000 4K IOPS @ 0.3ms

300,000 4K IOPS @ 0.5ms

500,000 4K IOPS @1.0ms

XCubeFAS Gen2, Product Line

XF3000-Series

(Intel® Xeon 6-Core, 16GB DDR4 RDIMM with Max 384GB Per Controller)



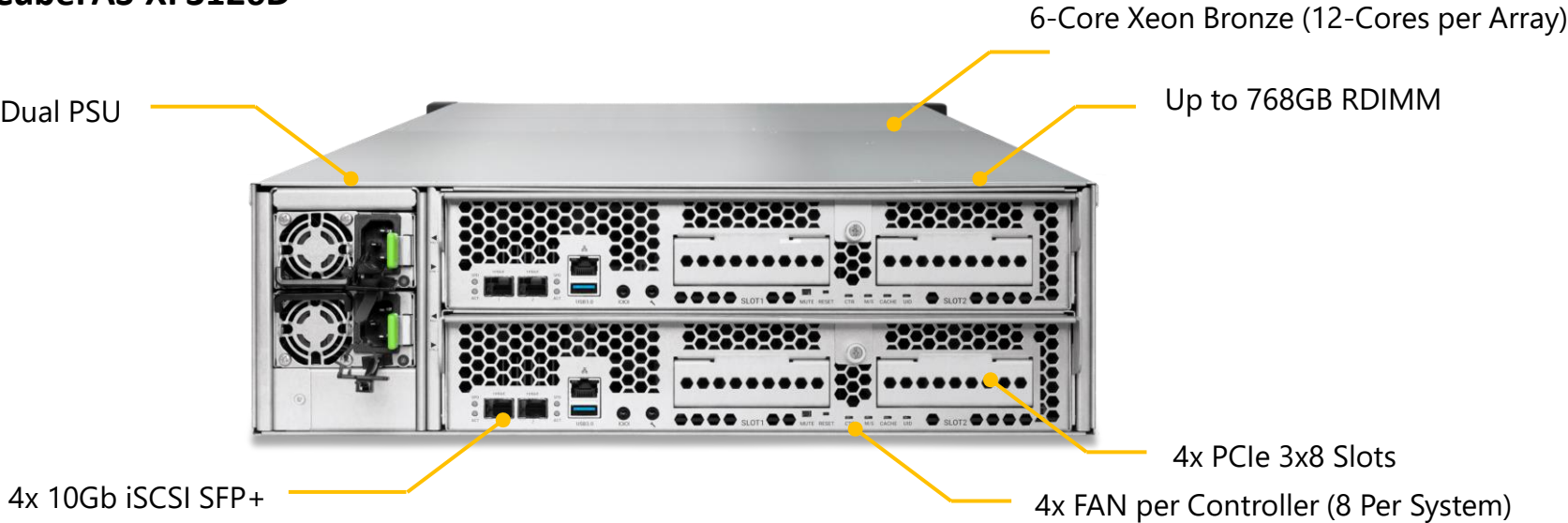
3U 26-bay

XF3126D

Flash-Based Management System



XCubeFAS XF3126D



Optional Host Cards

- | | | |
|-----------------|-------------------|---------------------|
| 2 port 32Gb FC | 2 port 25Gb SFP28 | 2 port 10GbE Base-T |
| 2/4 port 16G FC | 4 port 10GbE SFP+ | |

VDI Performance Reference (Monday Login Storm)

XCubeFAS
XF3126D

NetApp
C190

Normal Users
50-100
IOPs/user

1200 users
@1ms latency
(calculate 75 IOPs/user)

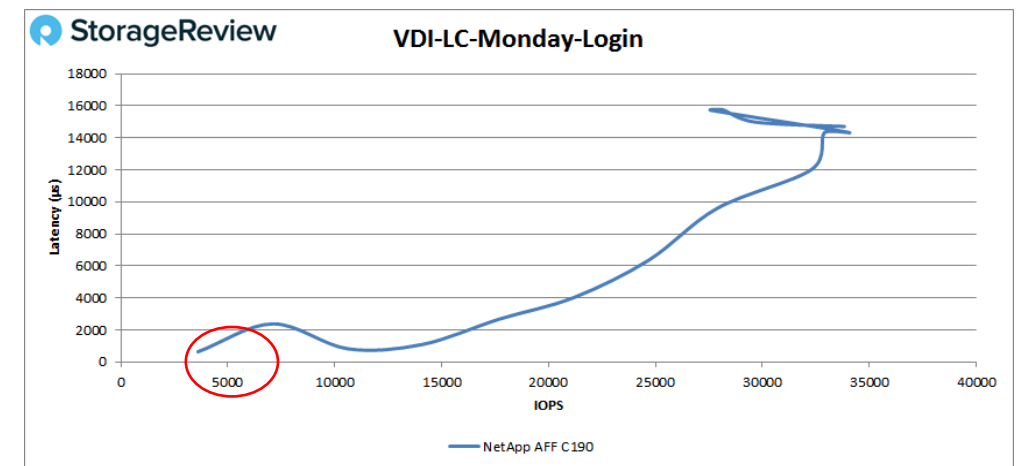
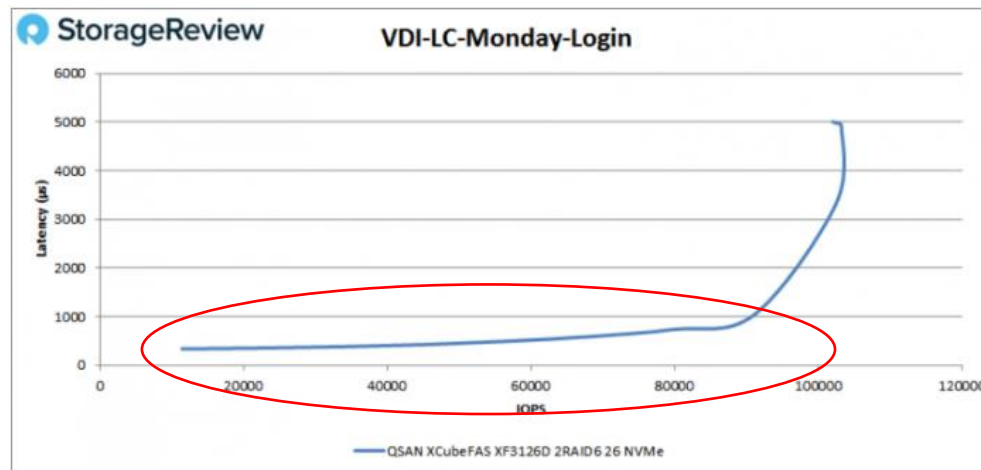
66 users
@1ms latency
(calculate 75 IOPs/user)

Performance
Reference
(StorageReview.com)

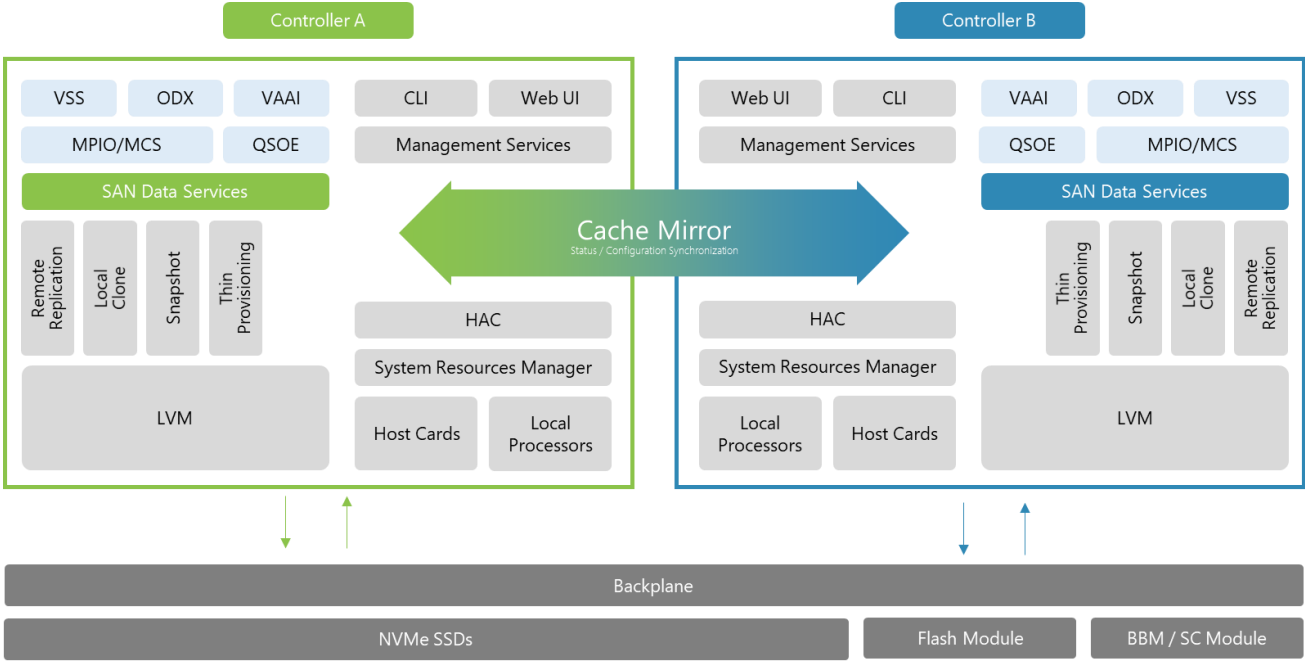
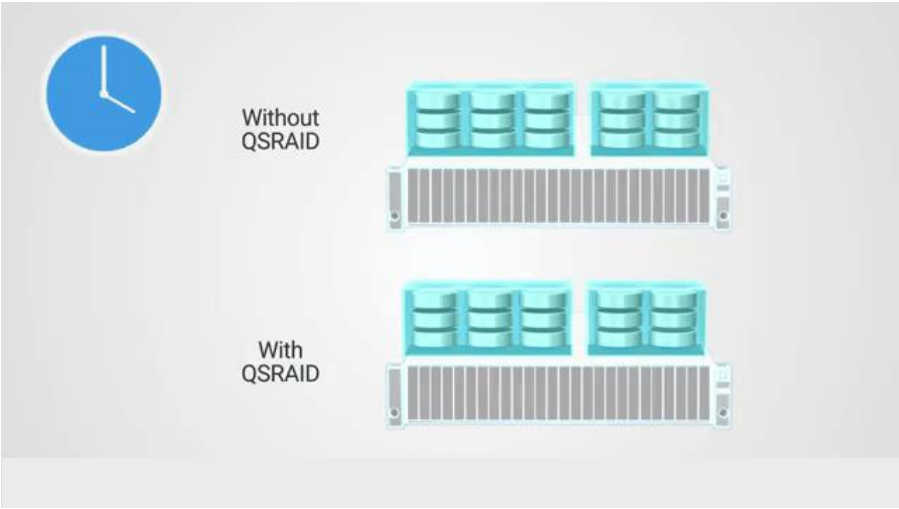
**90K @1ms
(RAID 6)**

**5K @1ms
(RAID 10)**

Under 1ms



XEVO 2.0



QSRAID
SSD Data Distribution

QSLife
SSD Life Monitoring

QReport
Historical Performance

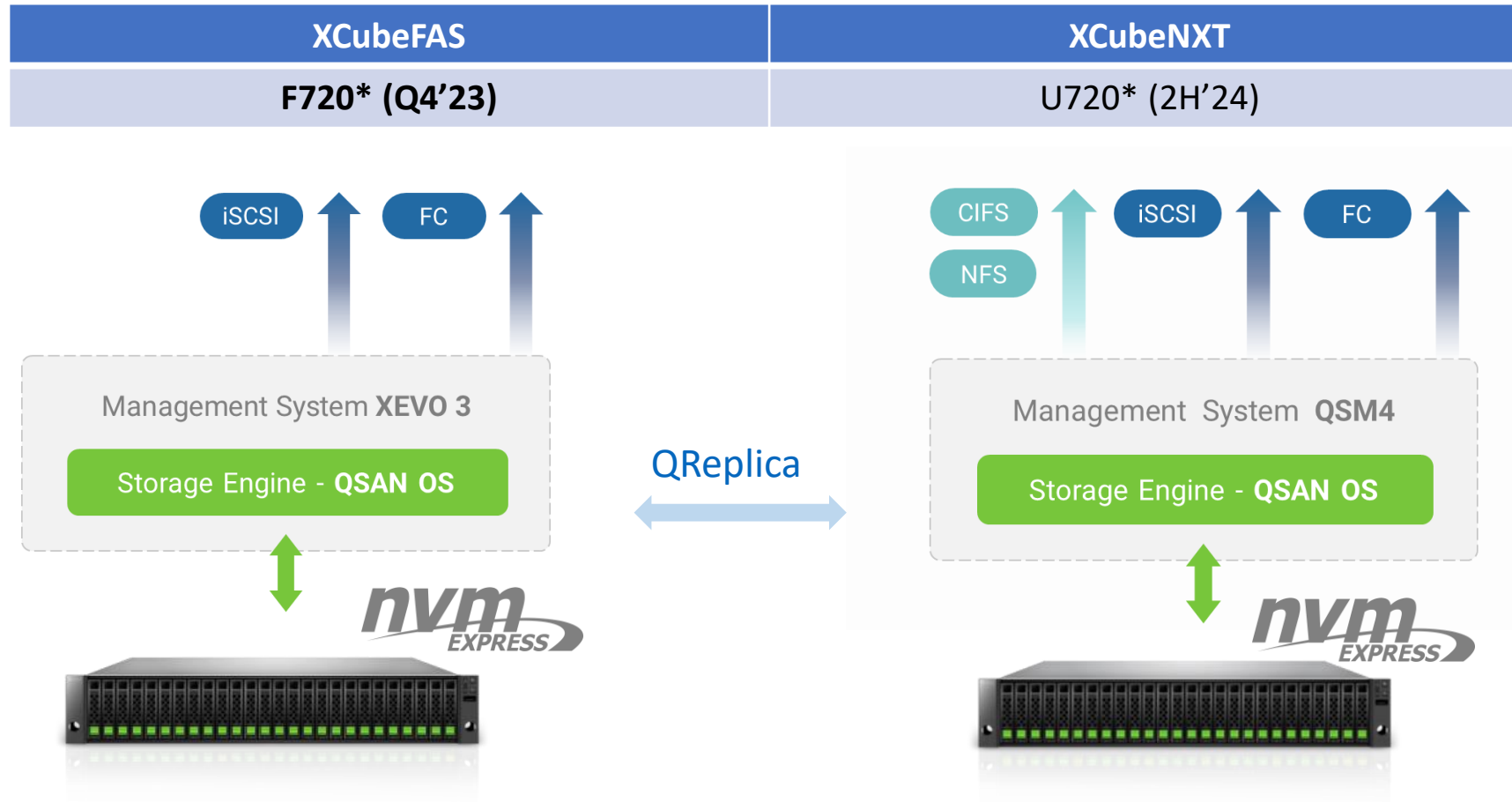
RAID EE
50% Faster RAID Rebuilds

QReplica
Remote Replication*

QSnap
16,000 Snapshots

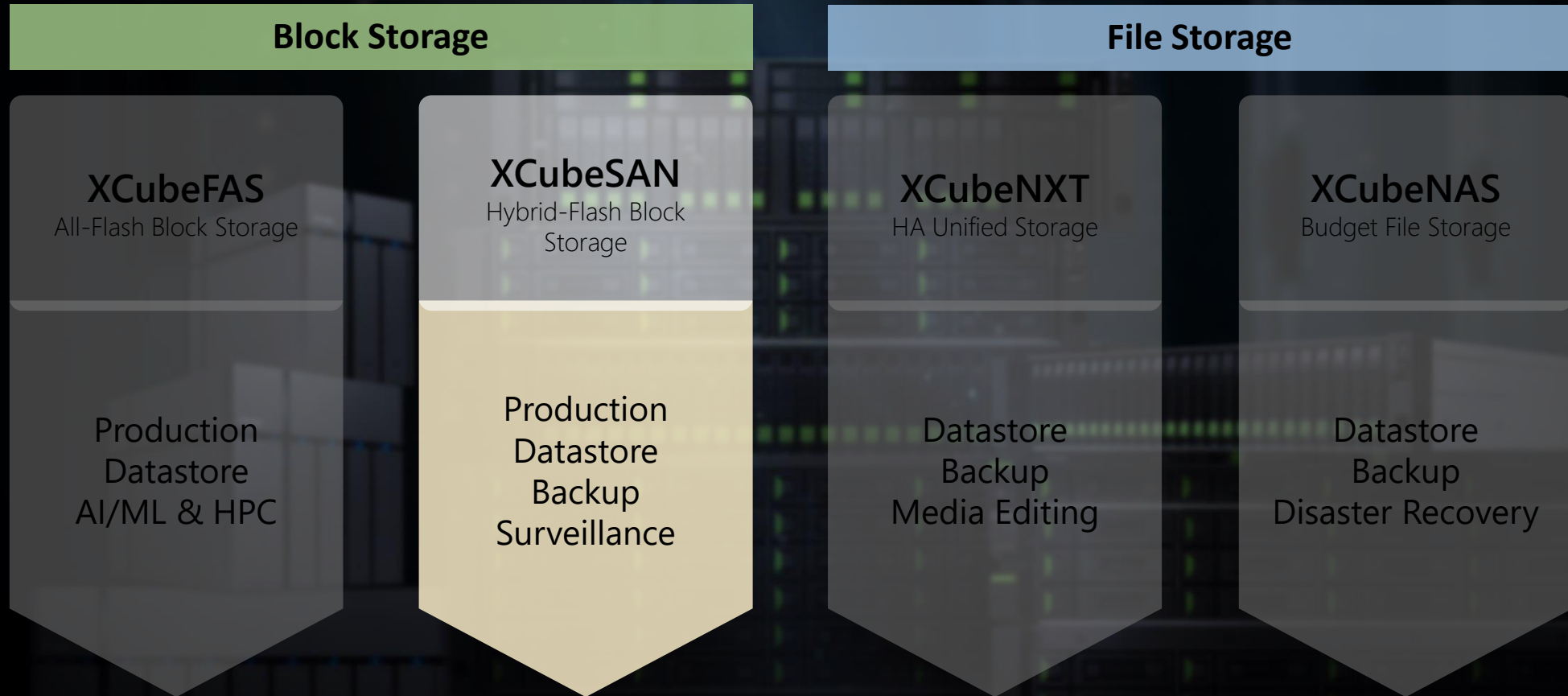
* Synchronous (optional license) /Asynchronous (built-in) block-level, differential volume backup

ЕДИНАЯ ПЛАТФОРМА (НОВЫЙ NVME AFA)



* Internal project name

XCube Platform – One Architecture



XCubeSAN Product Line

XS5300 Series
(Intel® Ice-Lake D, XEON 8/4 Cores, 16GB to 256GB DDR4 Memory Per Controller)

XS3300 Series
(Intel® Ice-Lake D, XEON 4 Cores, 8GB to 256GB DDR4 Memory Per Controller)

XD5300 Series
(Expansion unit, SAS 12G)

2U 12-bay

3U 16-bay

4U 24-bay

2U 26-bay

XS5312D
XS5312S

XS5316D
XS5316S

XS5324D
XS5324S

XS5326D
XS5326S

XS3312D
XS3312S

XS3316D
XS3316S

XS3324D
XS3324S

XS3326D
XS3326S

Expansion Enclosures

XD5312D
XD5312S

XD5316D
XD5316S

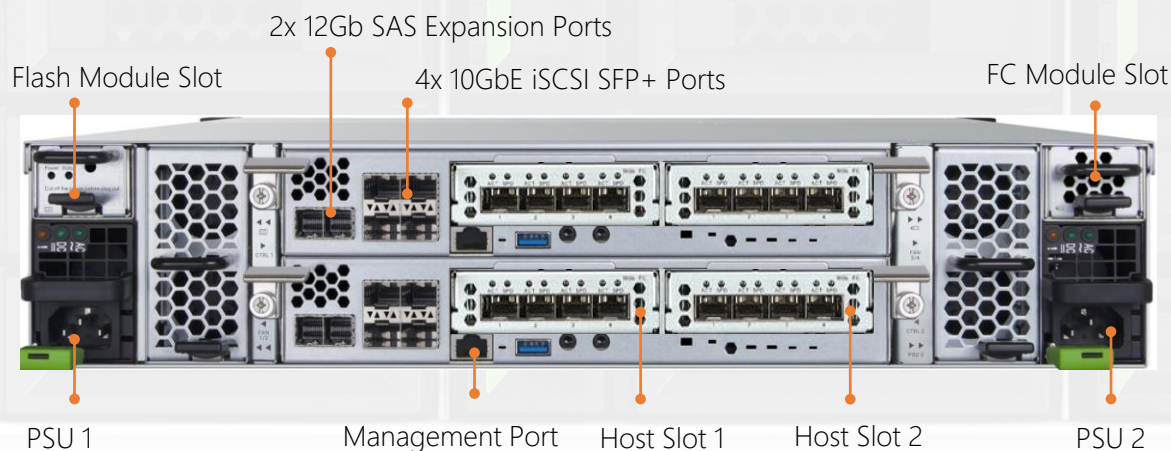
XD5324D
XD5324S

XD5326D
XD5326S

D: Dual Controller
S: Single Upgradable

LFF

SFF



Optional Host Cards and Accessories:

2/4 port 16Gb/32Gb FC

2 port 25Gb iSCSI

4 port 10GbE iSCSI SFP+

C2F 256G SC

Outstanding Scale-Up

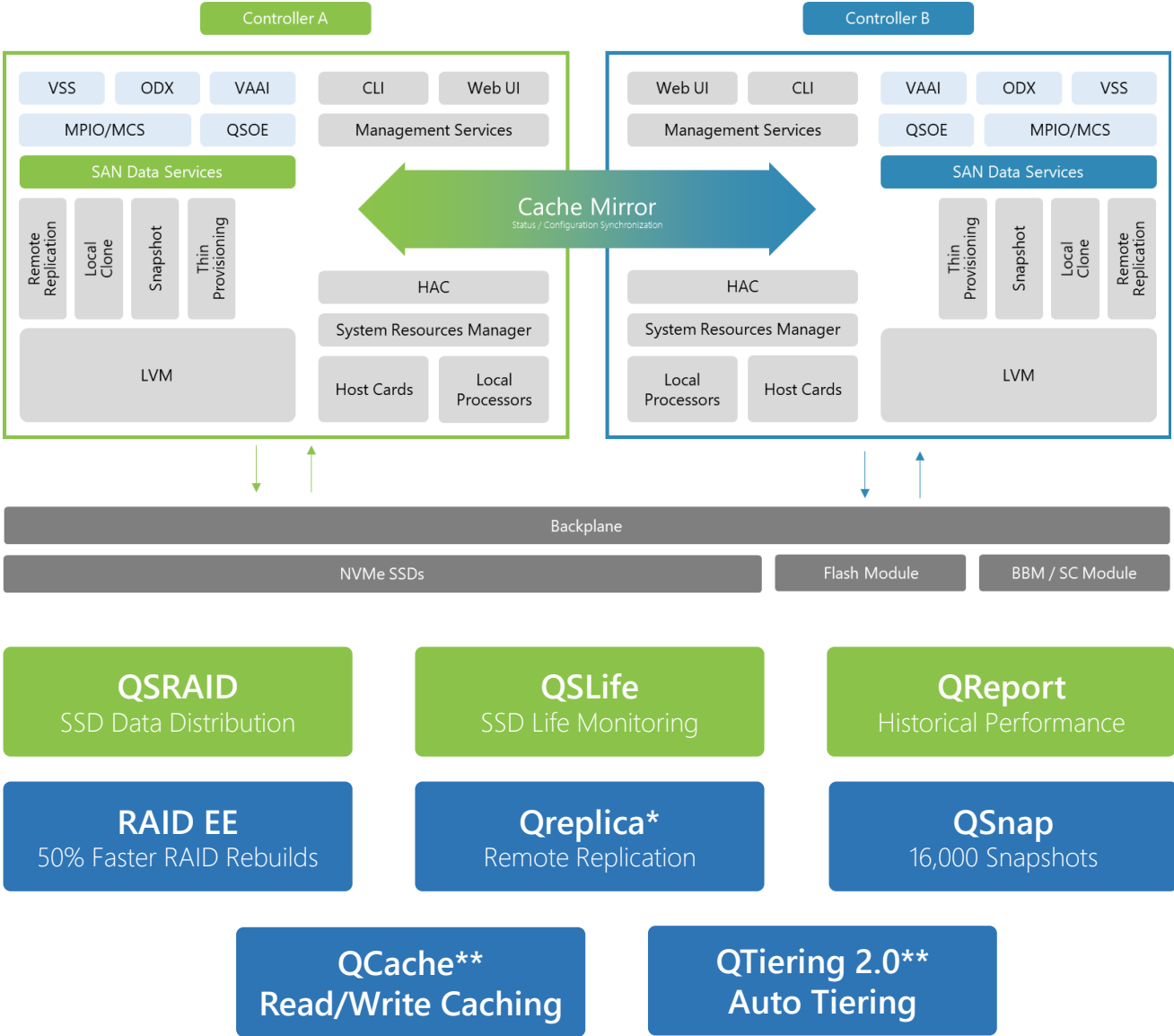
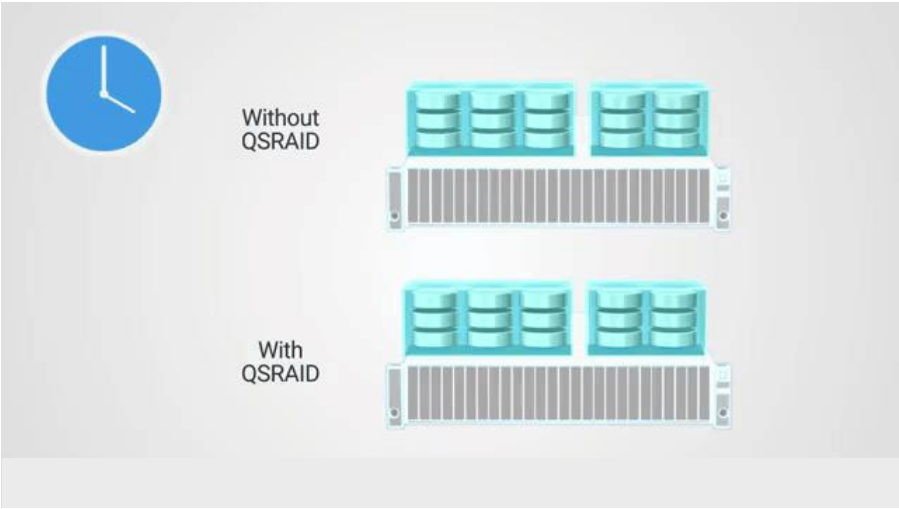
XS53/33



- Up to 20*XD5300 or 4*3rd-party JBODs
- Max. 546 drives
- Max. 16PB raw capacity



XEVO 2.0



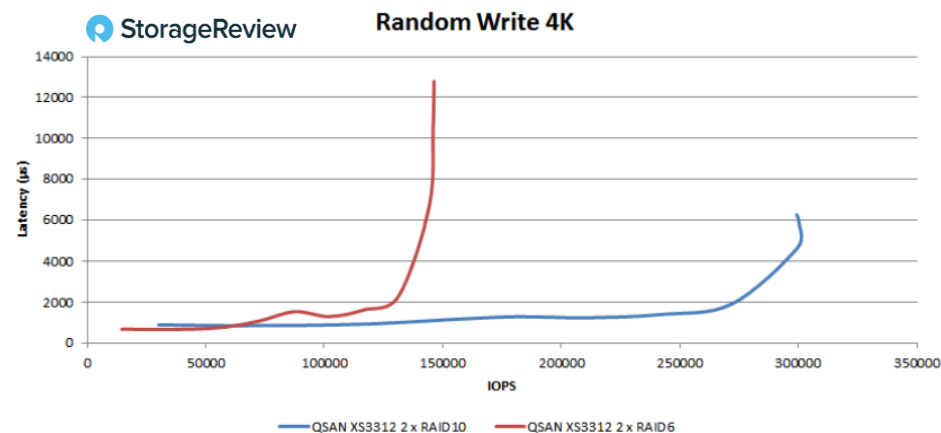
* Synchronous (optional license) /Asynchronous (built-in) block-level, differential volume backup

** Optional license

Futureproof Performance

QSAN XS3312D

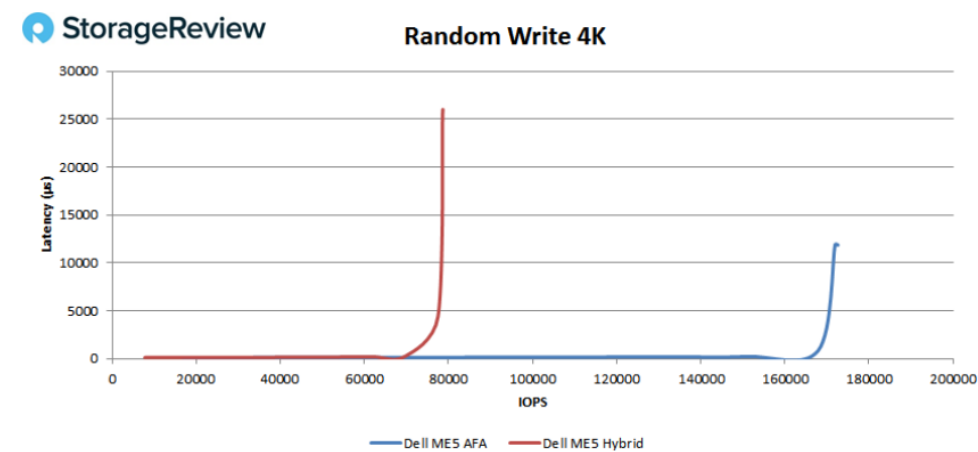
Max. 300K IOPS



<https://www.storagereview.com/review/qsan-xcubesan-xs3312d>

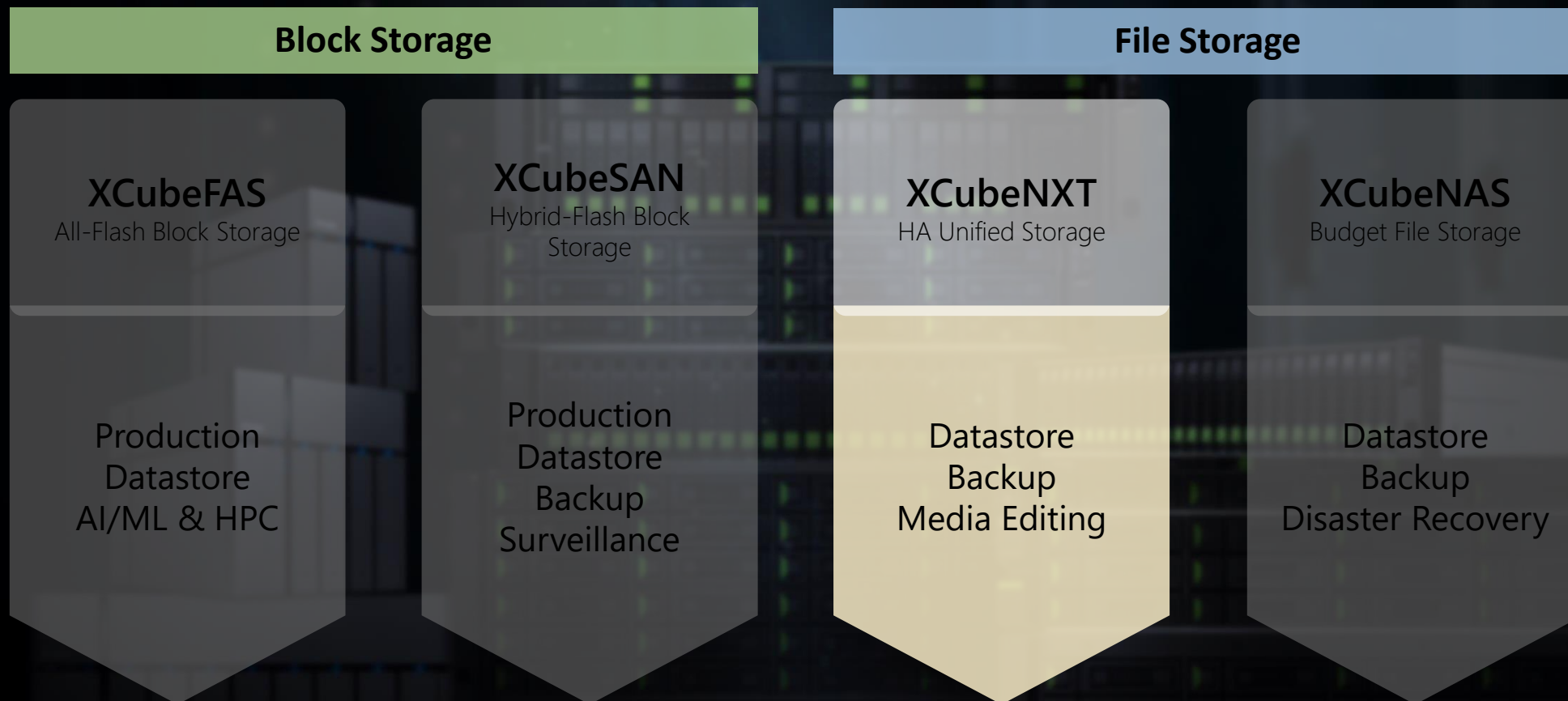
Dell EMC ME5

Max. 172.6K IOPS



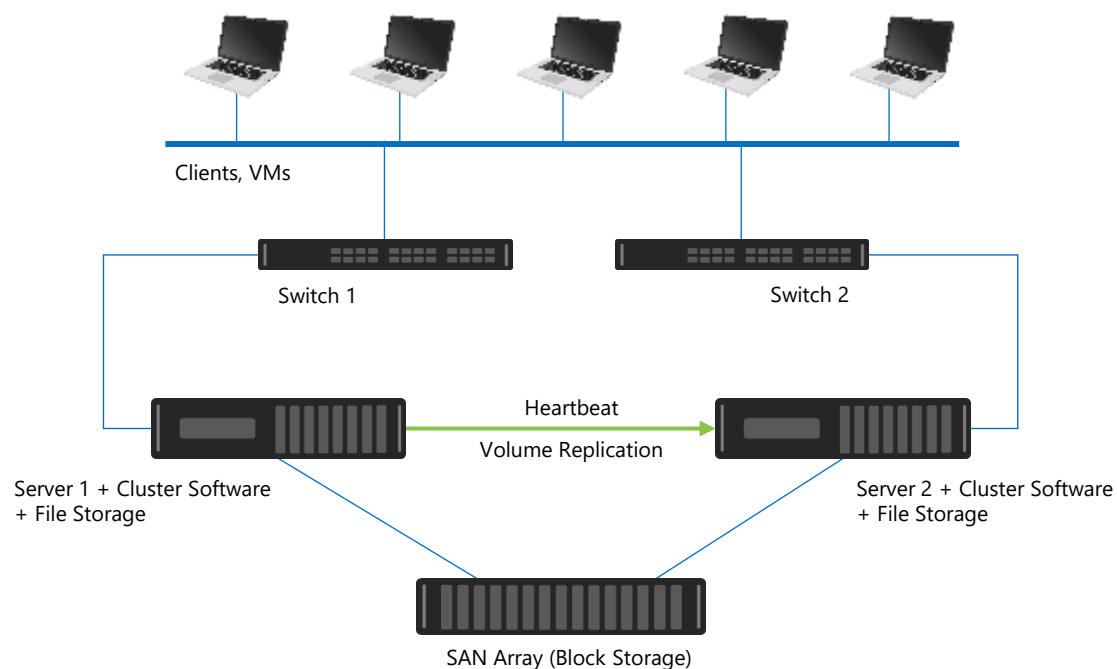
<https://www.storagereview.com/review/the-dell-powervault-me5-is-enabling-edge-applications>

XCube Platform – One Architecture

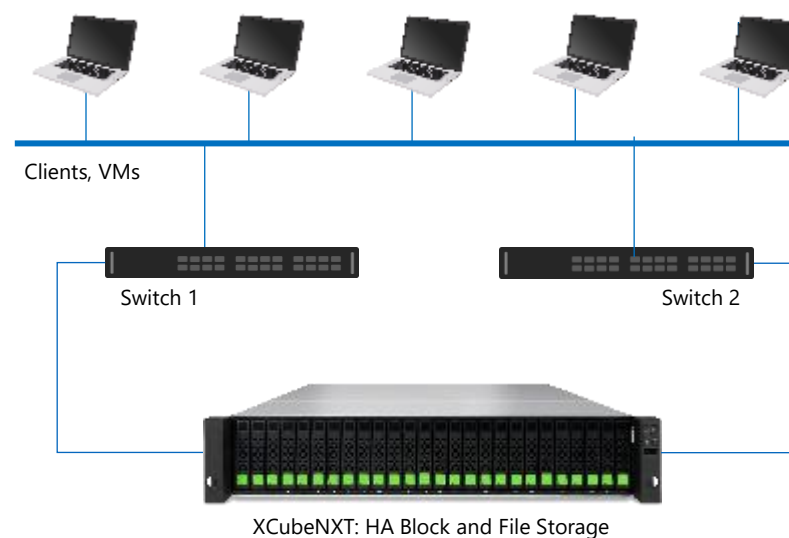


Единая платформа - Новое поколение NXT

Traditional HA Cluster



Unified Storage



Applications:

- Virtualization
- Data Center
- File Sharing
- Surveillance
- Backup and Archive

Линейка XCubeNXT

XN8100 Series
(Intel® Ice-Lake D, XEON 8/4 Cores, 16GB to 256GB DDR4 Memory Per Controller)

XN5100 Series
(Intel® Ice-Lake D, XEON 4 Cores, 8GB to 256GB DDR4 Memory Per Controller)

XD5300 Series
(Expansion unit, SAS 12G)

2U 12-bay

XN8112D
XN8112S

3U 16-bay

XN8116D
XN8116S

4U 24-bay

XN8124D
XN8124S

2U 26-bay

XN8126D
XN8126S

XN5112D
XN5112S

XN5116D
XN5116S

XN5124D
XN5124S

XN5126D
XN5126S

Expansion Enclosures

XD5312D
XD5312S

XD5316D
XD5316S

XD5324D
XD5324S

XD5326D
XD5326S

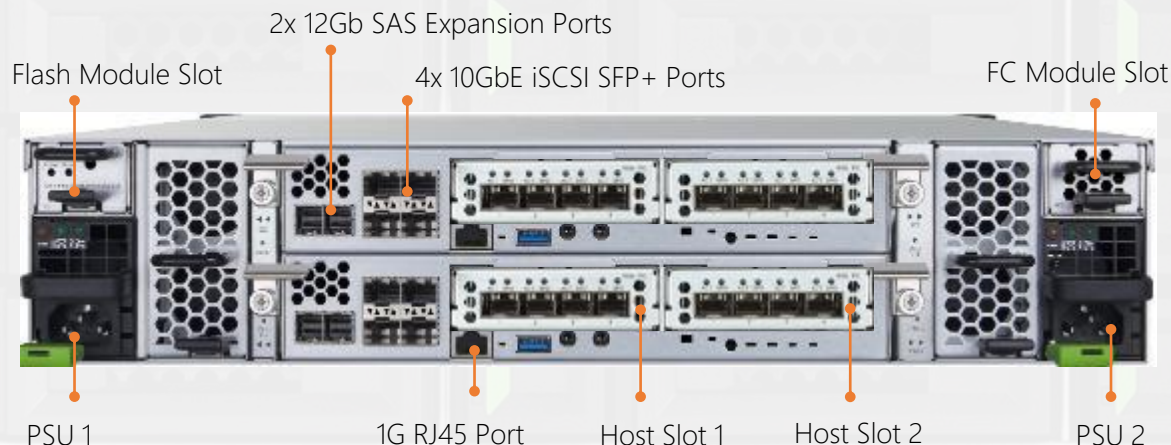
Random Read:
300K

Random Write:
180K

D: Dual Controller
S: Single Upgradable

LFF

SFF



Optional Host Cards and Accessories:

2/4 port 16Gb/32Gb FC*

2 port 25Gb iSCSI

4 port 10GbE iSCSI SFP+

C2F 256G SC

* Available in Q1, 2024

💡 Одна архитектура для различных форм-факторов: Экономия на управлении запасными частями!

QSAN

НОВОЕ ПОКОЛЕНИЕ NXT – СЕРИЯ XN8100 & XN5100

Серия XN8000



На контроллер:

- 2x 10GbE iSCSI (BASE-T) порт (встроено)
- 4x 10GbE SFP+ порт (опция)
- 2x/ 4x 16G FC порт (опция)
- 2x 25Gb SFP28 порт (опция)
- 2x 12Gb SAS Expansion порт (встроено)

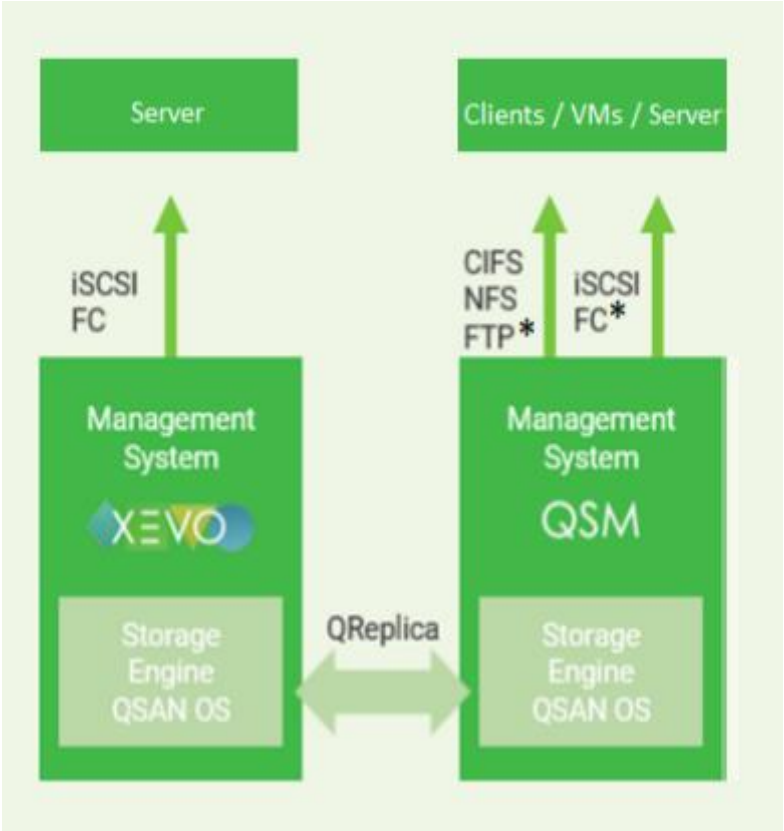
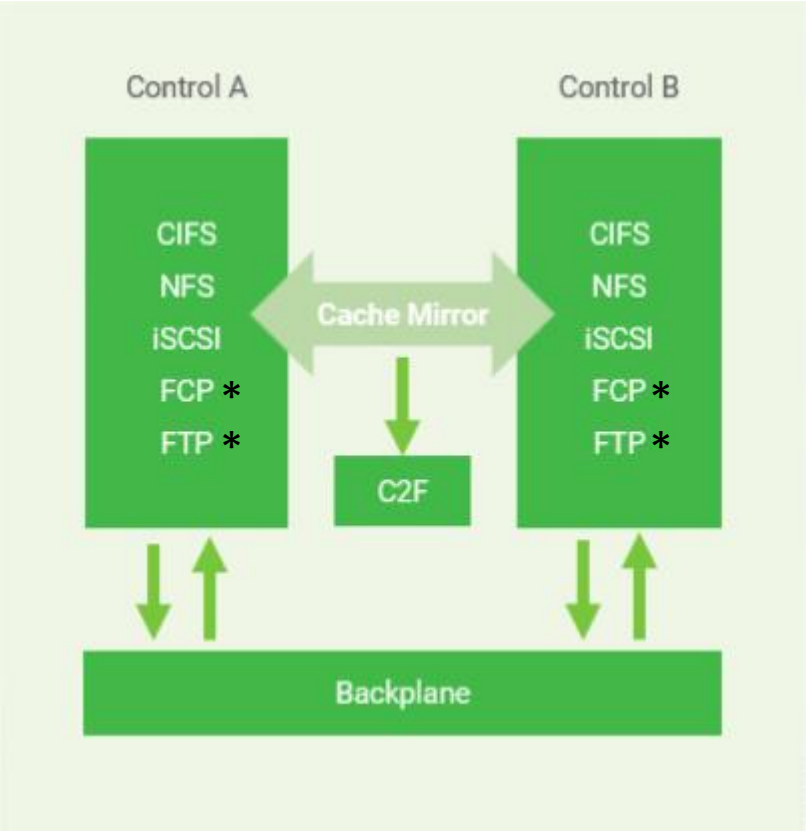
Серия XN8100 & XN5100



На контроллер:

- 4x 10GbE SFP+ порт (встроено)/ (опция)
- 4x 10GbE RJ45 порт (опция)
- 2x/ 4x 16G FC порт (опция)
- 2x 25Gb SFP28 порт (опция)
- 2x 32Gb SFP28 порт (опция)
- 2x 12Gb SAS Expansion Ports (встроено)

QSM 4



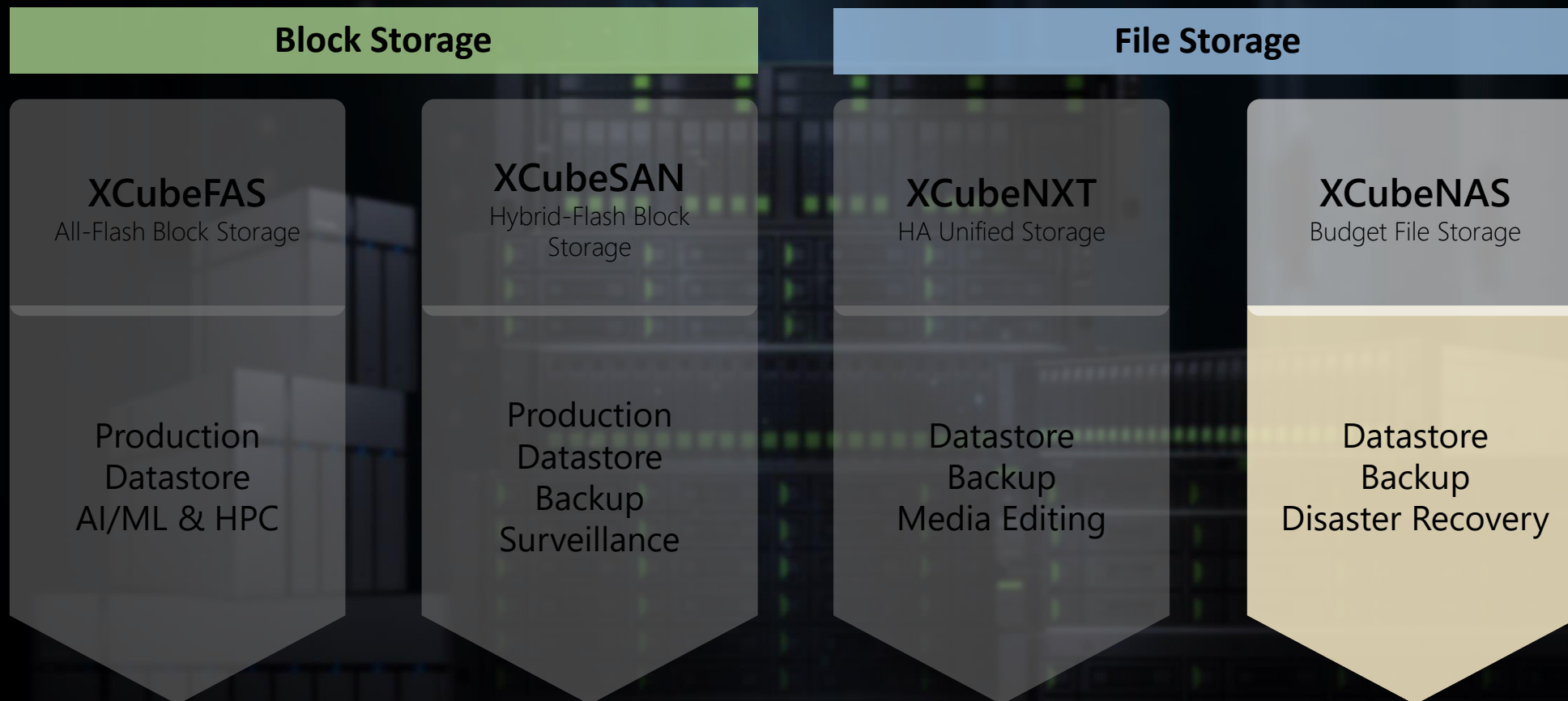
* Available in Q1, 2024

* Available in Q1, 2024

SED	WORM
Thin Provision	Compress/Dedup**
QReport Historical Performance	QSLife SSD Life Monitoring
QTiering 2.0 Auto Tiering	QCache Read/Write Caching
RAID EE 50% Faster RAID Rebuilds	QSnap 16,000 Snapshots
QReplica* Remote Replication	

* Synchronous (optional license) /Asynchronous (built-in) block-level, differential volume backup
** Optional license

XCube Platform – One Architecture



XCubeNAS Product Line



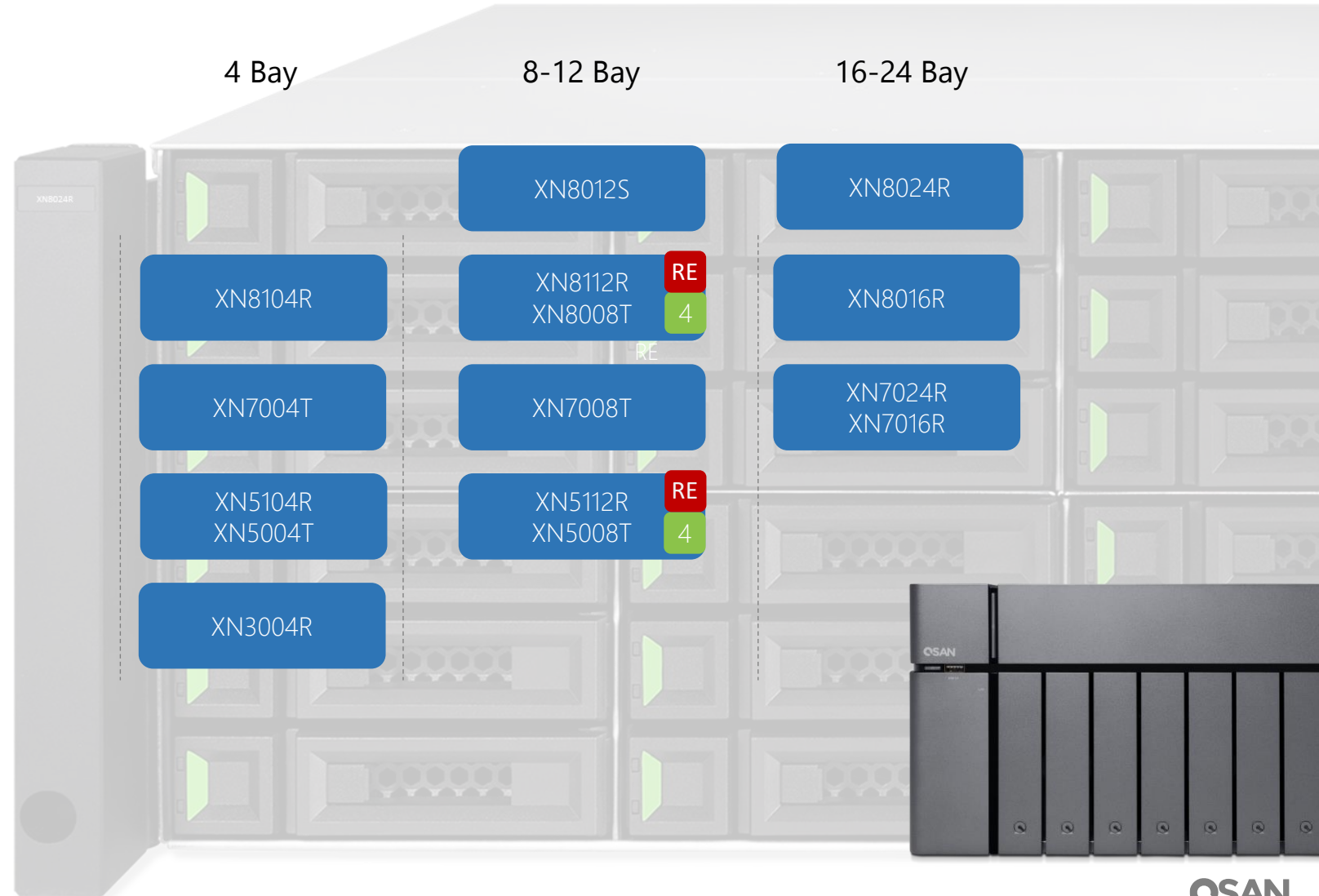
XN8000-Series
Rack Models: Intel Kaby Lake Xeon E3,
Tower Models: Intel Kaby Lake i5

XN7000-Series
Rack Models: Intel Kaby Lake i3
Tower Models: i3 / Pentium Q-Core

XN5000-Series
Rack Models: Intel Kaby Lake Celeron
Tower Models: Intel Kaby Lake Celeron

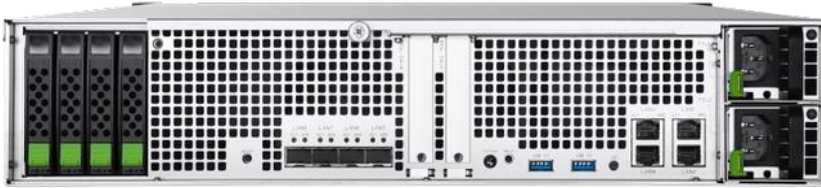
XN3000-Series
Tower Models: Intel Apollo Lake
Celeron

- N** Rack models with N-number of additional SSD bays
 - RE** 12 Bay Rack series without additional SSD Bays
 - T** All tower models are equipped with 1 additional SSD bay
- * All Rack models come with redundant PSUs



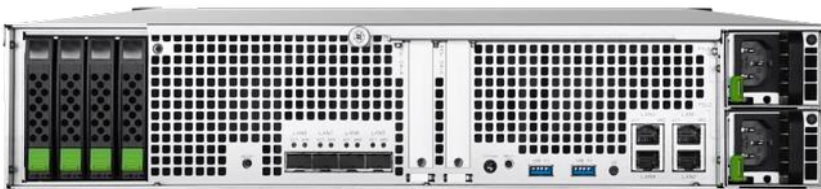
XCubeNAS 5100/8100

XN8112R



- 2 * M.2 NVMe SSD (встроено)
- 4 * U.2 NVMe SSD
- 4 * 10GbE SFP+
- 4 * 2.5GbE RJ45
- Max. 12 host connections
- Intel Xeon CPU
- ECC memory*

XN5112R



**ECC memory will protect system from a potential crash by correcting any errors in the data to avoid data corruption from hardware level.*

Integrated Storage Platform

QSAN and Western Digital

Form Factors
4U60, 4U102

Capacity
Up to 9PB

Interfaces
12x Mini SAS Ports

SSD Support
Up to 24 per JBOD



СПАСИБО ЗА ВАШЕ ВНИМАНИЕ.

For more information please visit: www.qsan.com

Stella Yang: stella.yang@qsa.com

